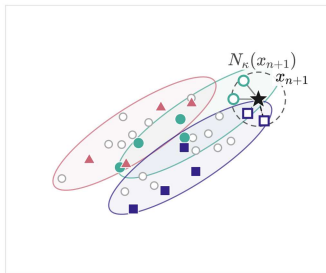
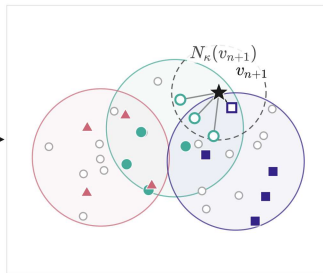
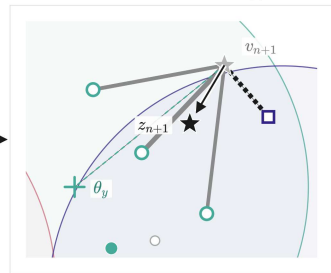
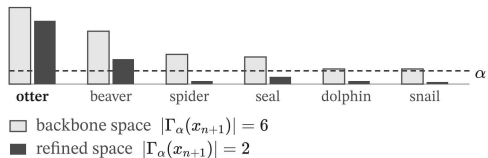
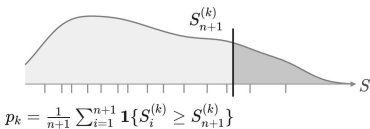


(a) backbone space $x = \phi(a)$ **(b) whitened** $v = \tilde{\Sigma}^{-1/2} V^{\top} (x - \mu_{\mathcal{U}})$ **(c) smoothed** $z \propto v + \sum_{u \in N_{\kappa}(v)} w_u u$ 

$$T_{\mathcal{U}} = T_{\text{smo}} \circ T_{\text{wht}} \circ T_{\text{trc}}$$

fitted on the unlabeled pool alone (Prop. 1)

(d) the conformal step

● ▲ ■ labeled sample $\mathcal{D}_n$

○ unlabeled pool $\mathcal{U}$

— --- neighbour edge, same class or other class